Hand detection using Generalized Hough Transform

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1 Abstract

Hand segmentation and pose estimation have been an area of major interest in the field of computer vision for decades now. The successful training and unmatched results of deep learning has shifted the focus from classical methods of computer vision and image processing to the hybrid of the two fields. While this provides faster solutions, the accuracy of classing vision algorithms is still not achieved through this method. In this paper we have explored the use of general hough transform with bag of visual codewords [1] to obtain acceptable image segmentation results.

2 Related work

For fast and reliable segmentation of the hands in Ego-vision random forest classifiers was used, as they are known to efficiently work even with large inputs. In [11] segmentation method (Simple Linear Iterative Clustering (SLIC) algorithm) was used which assign labels to super pixels. This allows a complexity reduction of the problem and also gives better spatial support for aggregating features that could belong to the same object.

Benefit from widely used Kinect camera, hand detection using both color and depth information has progressed rapidly. Hand region is extracted from depth image and the contour of the selected candidate is determined in the RGB image to improve the positional accuracy. And finally for tracking [9] used Generalized Hough Transform (GHT).

The Zivkovic background subtraction method is used in [12] before segmentation to separate background and foreground objects. This mask is then combined with the original image by means of a logical AND operation. Based on the colourcue hand location h(x,y), the hand is now segmented from the video frame. A combination of colour information, edge information, motion detection and skin probability is used to segment the hand, based on a combination of different color spaces is used to find certain parts of the hand contour.

In [15] prediction of 3D hand pose from single RGB images. PMD-Net comprised of two stages; in first stage the depth images were trained on depth-based network while in stage two RGB-based network was trained and their interface estimates the 3D pose from RGB network.

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In [3] they recognize gesture for which they need to segment and track three objects of interest: the face and the two hands. The segmentation of hand was done by segmentation of skin objects. They detect the skin by three useful features: color, motion and position. Color cue was used for skin detection. The motion cue for discriminating foreground from background pixels. And the position of object was found from the Kalman filter.

Aisha Urooj Khan et al [14] presented an in-depth study and discussion on the topic of hand-segmentation, the prevalent methods and their comparison in 2018. They contributed towards the issue by the using the two most frequently employed techniques to improve deep networks. First the fine-tuning of RefineNet was carried out which produced praise-worthy results for handsegmentation problem. Secondly three new datasets were also contributed to cater the occlusion and background problems in hand-segmentation.

Jonathan Tompson et al.[13] presented a 4-step real-time continuous pose recovery method in 2014 from a single depth image. First of all a decision forest classifier was used to segment out hands from the scenery followed by labeled dataset generation. This dataset was then passed through a convolutional neural network for feature extraction. The heat-map features were then used to obtain 3D pose information using reverse kinematics method.

HandVoxNet++, Voxel-based deep network [7] along with 3D and graph convolutions trained in supervised manner. This network relies on two hand shape representation. First the voxelised image is converted to heat map of joints (i.e. pose estimation) and uses 3D convolutions to capture the voxelized shape. Secondly this shape is passed through the graph-convolution-based mesh registration (i.e. GCN-meshReg) to preserve the mesh topology.

A novel algorithm is defined in [6] which uses Hough forest to estimate hand position from a single depth camera. It proposed to learn vote weights at leaf nodes of a forest in a way which minimizes average pose estimation error, so that ambiguous votes get suppressed during prediction stage. Hough forest is learned to map the local features of a set of Hough voting pixels to the probabilistic votes for the hand pose.

Jian Cheng et al. [2] tackled the issue of view variation and occlusion by proposing a view selection and fusion model for single-depth images using deep neural networks. In this model, the image is first converted to 3D point clouds and candidate virtual views are stationed uniformly on circular surface pivoted in hand point clouds. The cloud point is then extended as depth maps to candidate views which are used to predict the confidence via a light weight deep network. A pose estimation network is then used to estimate the hand pose with top confidence and fuse their results to give 3D hand-pose estimation. JoonKyu Park et al. [8] contributed to the occlusion problem as well by proposing a mesh estimation network called HandOccNet. The network is divided into two parts; feature-injecting transformer (FIT) and self-enhancing transformer (SET). FIT includes injecting information to the occluded region by finding information related to the secondary features in the primary features using correlation. The primary and secondary features are extracted using a spatial attention mechanism. SET then refines the output of the first part by using self-attention mechanism. MANO parameters are produced next via regressor and forwarded to MANO layer to obtain the 3D hand mesh.

Another approach for hand detection and pose estimation is the use of kinematic layer. Jan Wohlke , Shile Li , and Dongheui Lee [16] proposed such an approach where they extended the kinematic layer to make the hand pose parameters learn-able instead of rigid once-fed parameters. In this method, regression is applied on the input image and values for hand shape parameters such as palm shape and bone lengths are obtained. To decrease the variance in shape and size of input data, a series of appearance normalization networks is applied so that that images have an approximate normalized appearance.

A light weight model has been proposed by Nicholas Santavas et al [10] which uses a convolutional neural network in conjunction with a Self-attention model. The model used 1.9M parameters and has a size of 11Mbytes. This allows its deployment on low processing power devices and systems. A CNN based architecture Attention Augmented Inverted Bottleneck Block is used to obtain the keypoints' coordinates from single RGB image. This architecture is able to share collective knowledge among its layers resulting in a very effective model.

3 Deatiled Methodology

The Generalized Hough Transform (GHT) is a model based method for object localization, which has been introduced by Ballard [1] as an extension of standard Hough transform for line and circles. This method is robust against occlusion and noise and localize the object within an arbitrary shapes, which in our case is the localization of hand and different poses of hand. The generalized methodology of is shown in figure 1.

The localization procedure operate on two steps; first is making codebook of Codewords second is training phase or R-table construction and last stage is the recognition phase or accumulator array to vote for the center of hand.

3.1 Codebook of codewords:

In order to generate a codebook of different pose of hands, we use an approach inspired by the work of [5]. From training images features are extracted with the Harris interest point detector or descriptor [4]. These features are then clustered through K-means clustering algorithm to for clusters. The center of different

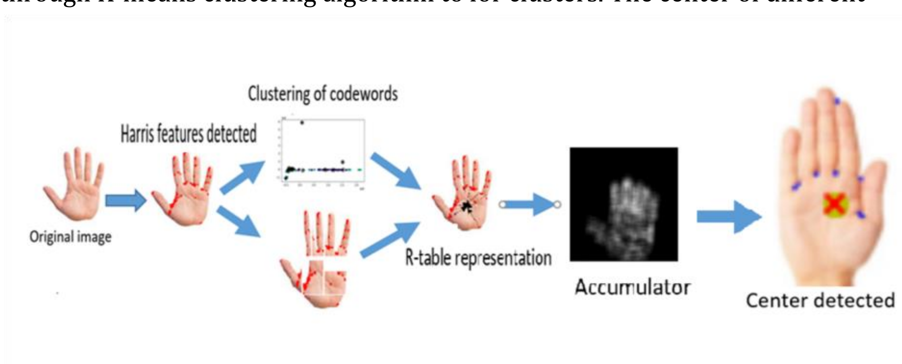


Figure 1: General methodology

cluster centroid form the codeword for that specific image. This codeword is then passed to the training phase to make phi-table.

3.2 Training phase

: Training phase operates by transforming the training image into the Hough space or also called as parametric space. Where each cell represent a certain model transformation. In our case only translation vectors are consider, a full affine transformation would be conceivable, but is computationally expensive. For localization procedure, Codebook representing the features of the hands is needed. This model consist of codeword relative to reference point and the gradient direction at each codeword. Then using the codeword and their corresponding gradient information a displacement vector is find out. This displacement vectors are stored in the R-table where they are grouped according to their gradient direction for faster access during voting procedure.

3.3 Recognition phase:

To perform localization, the edge map is computed, using canny edge detector. For each edge point, the model points with similar gradient direction are obtained from the R-table bin. For each entry in the R-table we compute the x and y using equations as follows and votes in the accumulator array is incremented. In the end, the accumulator array with maximum votes is returned as the most likely position or center of the hand.

4 Experiments:

We perform experiments on three types of hand images which are as follows:

1. Single hand dataset of sign.
2. Multiple hands in single picture.
3. Hand images with background other than white color.

4.1 Single hand dataset of sign:

In training phase we pass through the image of single hand as shown in figure 2 it makes the codewords for that images. The gradient were calculated for a particular codeword and its displacement vector was saved in the r-table. From the r-table accumulator array of test image was made. In our code the accumulator array does not show up in single hand images.the results are shown in result section (6)

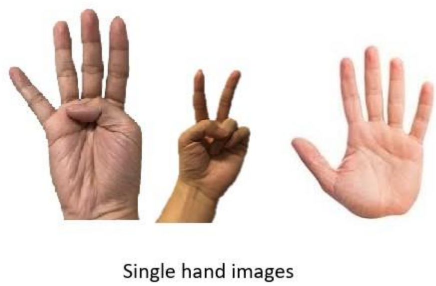


Figure 2: Single Hand Images

4.2 Multiple hands in single picture:

For multiple hands in single image the same procedure for training and recognition were used and the results were quiet promising. We gave the training phase single hand image as shown in the figure 3 from the same image and it correctly detect the center of that particular hand. Moreover in this case the Hough also made the accumulator array which shows the hand center.

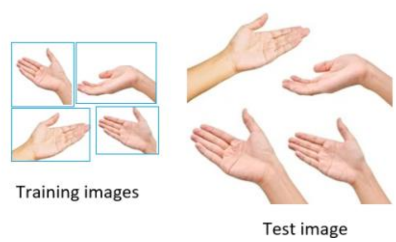


Figure 3: Multiple hands in single picture

4.3 Hand images with background other than white color:

For this kind of images the background was a problem because all the features of the background was detected using Harris interest point for making codebook. So we have to subtract the background from the image to get the results.

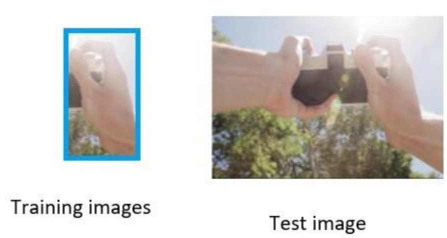


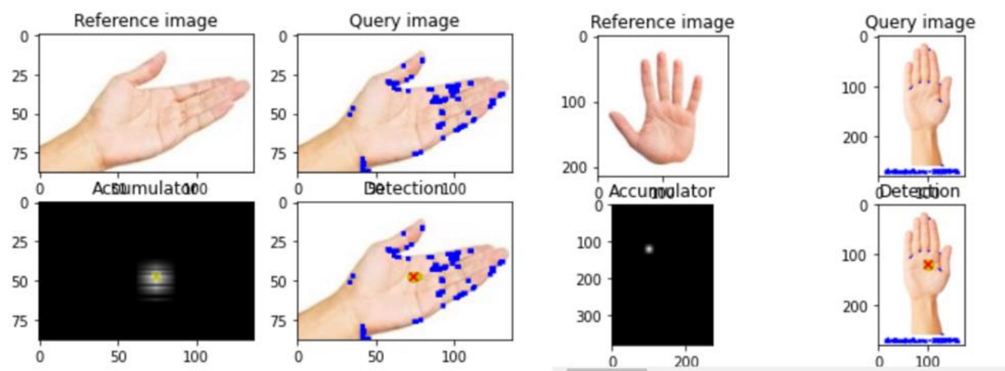
Figure 4: Multiple hands with colored background

5 Results:

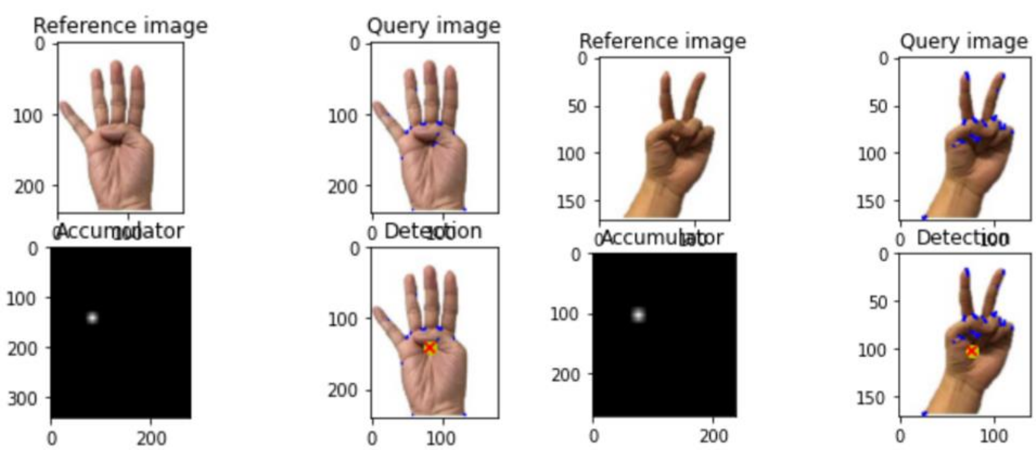
Step-wise experimentation was carried out as menyioned in the above section and following results were obtained.

5.1 Single hand dataset of sign:

For single hand images, feature descriptors were correctly located and using the bag of codewords, the center point of hand in those images were successfully located.Following are the result we get for such data images.

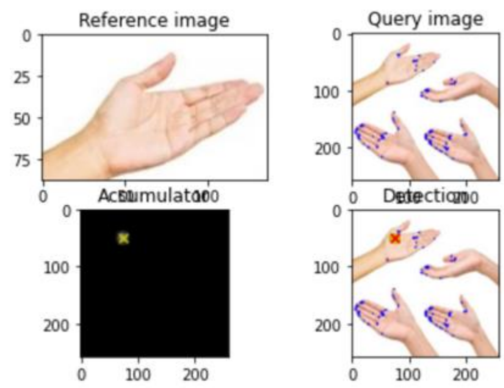


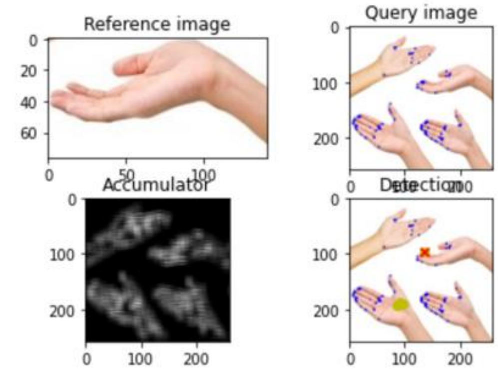
The below images are from sign language dataset. Experiments on this dataset were fully successful and 100% results were obtained for single hand segmentation.

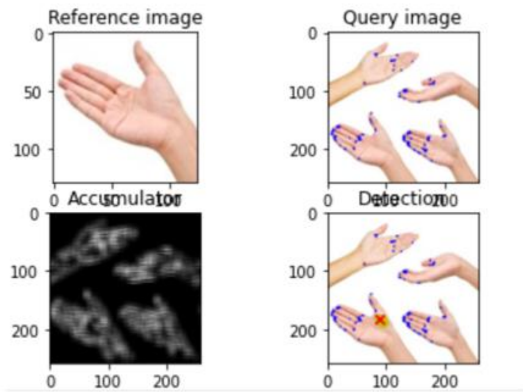


5.2 Multiple hands in single picture:

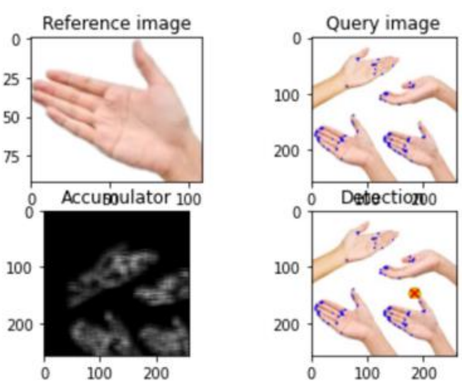
The following correct detection of hand are as follows the yellow circle are the top 5 result while the red cross is the top result.

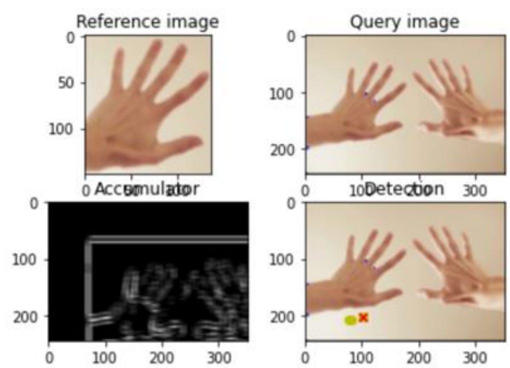






Sometimes it give ambiguous result as shown due to background colour.

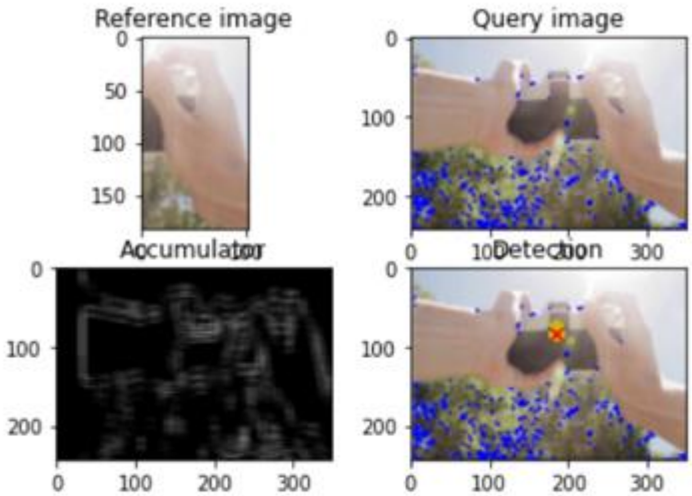




5.3 Hand images with background other than white color:

Because the features of training image was detected from background so the center of hand was not properly detected.

This was the image after features were detected.



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